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Review

Cooling Techniques for Enhanced Efficiency of Photovoltaic Panels—Comparative Analysis with Environmental and Economic Insights

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Abstract: Photovoltaic panels play a pivotal role in the renewable energy sector, serving as a crucial component for generating environmentally friendly electricity from sunlight. However, a persistent challenge lies in the adverse effects of rising temperatures resulting from prolonged exposure to solar radiation. Consequently, this elevated temperature hinders the efficiency of photovoltaic panels and reduces power production, primarily due to changes in semiconductor properties within the solar cells. Given the depletion of limited fossil fuel resources and the urgent need to reduce carbon gas emissions, scientists and researchers are actively exploring innovative strategies to enhance photovoltaic panel efficiency through advanced cooling methods. This paper conducts a comprehensive review of various cooling technologies employed to enhance the performance of PV panels, encompassing water-based, air-based, and phase-change materials, alongside novel cooling approaches. This study collects and assesses data from recent studies on cooling the PV panel, considering both environmental and economic factors, illustrating the importance of cooling methods on photovoltaic panel efficiency. Among the investigated cooling methods, the thermoelectric cooling method emerges as a promising solution, demonstrating noteworthy improvements in energy efficiency and a positive environmental footprint while maintaining economic viability. As future work, studies should be made at the level of different periods of time throughout the years and for longer periods. This research contributes to the ongoing effort to identify effective cooling strategies, ultimately advancing electricity generation from photovoltaic panels and promoting the adoption of sustainable energy systems.

Keywords: photovoltaic panels; cooling techniques; environmental and economic study; future recommendations

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1. Introduction

The continued population growth has resulted in the need for more energy resources to satisfy different sectors of life [1–4]. Further, the continued use of fossil fuels has led to depletion of resources and increases in price and CO_2 in the atmosphere. Therefore, current research focuses on finding alternative solutions through renewable energy resources [5,6] and heat recovery systems [7–10].

Solar energy forms an important factor in renewable energy resources, mainly through photovoltaic (PV) panels. Solar-energy-based PVs constitute a widely used technology in

modern life based on the principle of converting sunlight into electricity through semiconductor materials. This technology enabled a great leap forward in the world of renewable energy resources due to its environmental impact on the reduction in CO_2 emissions, its fast payback period, and its long maintenance period (every 25–30 years). However, the need for innovative installation techniques on modern roofs, the high prices, and the low power generation on rainy days are obstacles to the installation of this technology.

The main obstacle in this technology is its low efficiency due to high temperatures. The constant contact of sun rays at the surface of the PV panel increases its temperature, thus decreasing its efficiency and output power. It was found that the efficiency of crystalline silicon solar cells falls by 0.45–0.6% for every 1 °C rise above STC (standard test conditions) in solar cell temperatures and varies according to the type of cell [11].

To increase the efficiency and the affordability of the panels, different approaches were recorded in the trial to reduce solar cell temperatures. In the literature, four cooling techniques are demonstrated with their different methods. The first technique is using passive and active cooling methods of water. The second cooling technique is the use of free and forced convection of air. The third cooling technique is the use of phase-change materials (PCM) to absorb the excess of heat produced by the PV panel. Then the last cooling technique is a sum of uncategorized and modern methods.

Table 1 portrays a collection of recent studies on different cooling techniques of photovoltaic panels using novel approaches. The studies cover research and review articles.

Table 1. Recent research conducted on cooling PV panels using different novel methods.

Objective	Methodology	Outcomes	References
Review on photovoltaic–thermal collector technology and advances in thermally driven cycles for PVT collectors.	Literature review on PVT collector types, discussion of cooling solar systems, their limitations, and future recommendations.	Electrical and thermal efficiency enhancement up to 11% and 22.02% maximum, respectively. The minimum payback period for PVT systems is 8.45–9.3 years.	Jiao et al. [12]
A comprehensive review of different cooling techniques used for concentrated PV cells.	Literature review on cooling CPV cell categories, discussion of CPV cooling systems, mentioning their advantages and disadvantages, and future recommendations.	Agreement between experimental and numerical results on enhancing the efficiency.	Ibrahim et al. [13]
Review on state-of-the-art photovoltaic thermal collectors and their abilities to increase energy production and CO_2 reduction.	Literature review on PVT systems, classification, discussion on performance enhancement, applications, and future recommendations.	The curve of emissions (Remap) could be reduced by 16% by 2030 if PV technology was used.	Herrando et al. [14]
Review on water-based PV systems and factors affecting them.	Literature review on cooling PV panels methods, classification of water-based cooling methods, discussion and analysis of these methods in a statistical manner.	Water-based cooling was shown to be effective in unused water spaces and has the potential to increase PV performance.	Ghosh [15]
A comprehensive review on cooling PV systems.	Literature discussing the different factors affecting the solar systems. Providing discussions on temperature mitigation strategies and cooling methods.	Discusses power plant performance, performance-affecting factors, and solutions to reduce the effect of those factors.	Aslam et al. [16]
Review on photovoltaic thermal systems in buildings and their application in heating, cooling, and power generation.	Literature discussing PVT systems and their integration into buildings, state-of-the-art systems designed for cooling, heating, and power production, and their limitations.	Hybrid systems showed the best performance, highlighting that PVT technology is still under development.	Herrando et al. [17]
Review on PV cooling technologies and their environmental impacts.	Literature review on PV technology, cooling techniques, advances in cooling technology, and future recommendations.	Air cooling was found to be cost-effective and simple, liquid cooling was found to be efficient but expensive, PCM cooling was found to enhance thermal efficiency but bulky, and nanomaterial was found to be efficient but expensive.	Hajjaj et al. [18]
Review on PV cooling using floating and solar tracking systems.	Literature review on PV panels, cooling methodologies, solar tracking, floating PV systems, and future recommendations.	Solar tracking and floating PV systems were found to reduce land usage and increase PV performance.	Hammoumi et al. [19]
Review of PV cooling technologies and their abilities in temperature reduction and power enhancement.	Literature review on cooling methods, discussing experimental studies and cooling systems limitations.	PCM combined with nanoparticles was found to be the most effective in cooling compared to water and air-based systems.	Sheik et al. [20]

Table 1. *Cont.*

Objective	Methodology	Outcomes	References
Review on photovoltaic thermal systems combined with PCM cooling.	A literature review was conducted about different cooling methods, traditional and advanced PV-T with PCM systems, and their potential, analyzing their performance, mentioning the challenges, and future recommendations.	Combined PV-T PCM systems are owed a 3–5% increase in electrical efficiency, 20–30% in thermal efficiency, and cost reduction by 15–20% with a payback period of less than 6 years compared to PV-T systems without PCM.	Cui et al. [21]
Review on PV passive cooling techniques.	Literature review on passive PV cooling methods, discussing the passive cooling methods while mentioning the unsolved challenges, and recommending future work.	Natural air ventilation and floatovoltaics cooling systems were found to be the most effective among the other passive cooling methods.	Mahdavi et al. [22]
Review on nano-based cooling techniques.	Literature review on nano-based PV cooling, classifying and discussing each method, and proposing designs and future recommendations.	Compared to conventional cooling methods, the hybrid nano-based cooling method could reduce PV's surface temperature by up to 16 °C and increase electrical efficiency by up to 50%.	Kandeal et al. [23]
Review on and comparison of solar tracking systems.	Literature review on PV panels, and solar tracking systems while categorizing them and focusing on dual-axis tracking, giving insights and future recommendations.	Dual-axis solar tracking systems were found to be more efficient at the level of PVs' performance compared to single-axis tracking systems and fixed systems.	Awasthi et al. [24]

In summary, this review paper aims to comprehensively explore various aspects of photovoltaic cooling methods. Most research concentrates on discussing specific cooling systems or evaluating them from a performance perspective, including photovoltaic–thermal collectors, concentrated PV cells, PVT systems in buildings, environmental impacts of cooling technologies, and various cooling methods such as air cooling, water-based systems, phase-change materials, and passive cooling techniques. The manuscript's novelty lies in its discussion of different technologies used in cooling PV panels while providing insights into the economic and environmental benefits of each cooling method.

This comprehensive review paper takes a unique and methodical approach to exploring various cooling methods for photovoltaic panels, distinguishing itself from previous research that often narrowly focused on specific systems or performance aspects. The goal is to provide a thorough and current analysis of advanced cooling technologies for solar systems, shedding light on both their economic and environmental benefits. Covering a diverse array of topics, from photovoltaic–thermal collectors to concentrated PV cells, the review showcases advancements in electrical and thermal efficiency, resulting in significant reductions in payback periods. This study emphasizes the critical role that cooling methods play in enhancing the sustainability and efficiency of PV systems. Noteworthy findings include the effectiveness of hybrid systems, thermoelectric, phase-change materials, and nano-based cooling methods in improving overall PV performance. Through this systematic categorization and assessment, coupled with insightful economic and environmental considerations, this research contributes valuable recommendations for future studies and advances in the realm of PV cooling methods, making a substantial contribution to the field.

The manuscripts mentioned in Table 1 provides valuable insights on future work and limitations that should be addressed that could be conducted in this field such as

- PVT collectors should take into consideration the available space of installation.
- Heat pipe PVT collectors are better in cooling than PVT collectors with refrigerants. However, their manufacturing and installation could be challenging.
- BIPVT collectors reduce the use of fossil fuels through offering savings at the level of electricity production and the materials that could be used.
- The PCM selection to be used for cooling could be challenging and depend on many factors. Studies should be performed at the level of the PCM to select the optimal one for this study.
- Pulsating flow for CPV cooling was found to increase the PV performance. It is suggested that this could be overcome through experimentation with the vibrations that come with pulsating flow for CPV collectors.

- It is suggested to study CPV cooling with the integration of porous media, PCM, or nanofluids.
- Building artificial intelligence devices to remove accumulated dust on PV panels as a means of cleaning and increasing efficiency.
- Despite the amount of research conducted in this field, more research needs to be performed to cover the different aspects of PV deterioration.

2. Theoretical Background

2.1. Principle

The phenomenon of photovoltaic energy was first discovered by Edmund Bequerel. The principle behind it is that when a photon reaches a semiconductor, two conductors are created: the free electron and the electron hole through rejection of the electrons by the negative transitional surface of the polarity. The released electrons flow to the upper layer. In the bottom layer, the electrons are transferred from one atom to the other in order to fill the empty spaces. Free electrons are conducted from the upper layer into the electric field, where the solar cell is located. The constant contact of sunlight on the surface of the solar panel ensures the continuity of electricity generation.

2.2. Parameters Affecting Panel Efficiency

Scientists and engineers found through experimental and numerical studies that different parameters other than panel temperature would affect its efficiency. Jathar et al. [25] reviewed the different environmental factors affecting PV panel efficiency. Environmental factors affecting panel efficiency are shown in Figure 1.

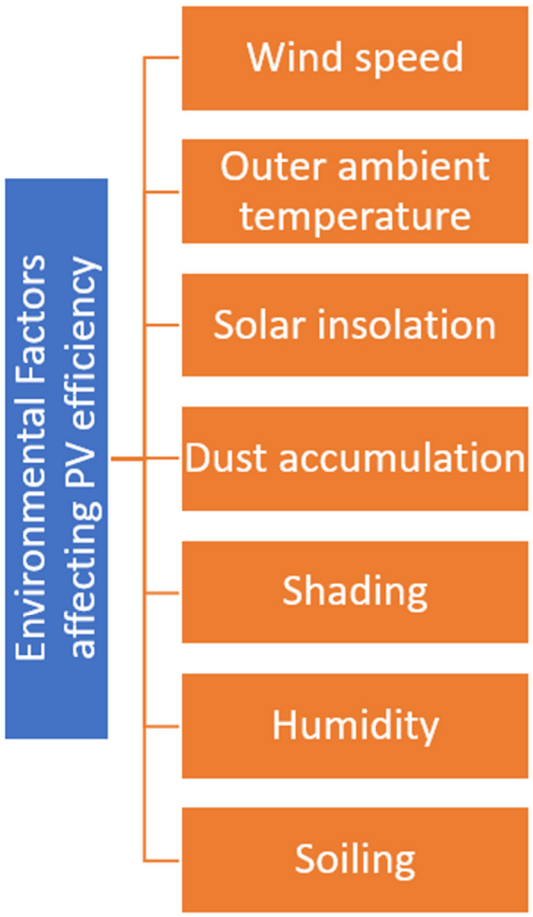


Figure 1. Environmental factors affecting the efficiency of PV panels [25].

2.3. Effect of Temperature on Panel Efficiency

Among all the mentioned parameters in Figure 1, temperature is dominant in efficiency deterioration. A PV panel absorbs approximately 80% of the incident radiation, but not all of it is converted into electricity. A definite range of wavelengths can be converted into electricity and all the others are converted into heat [26]. The remainder unconverted wavelengths can increase the solar cell temperature above the atmospheric temperature [27].

The current literature has proven the decrease in temperature coefficients (such as PV voltage and open-circuit current) with the increase in temperature [28]. Chander et al. [29] carried out an experimental study employing a solar cell simulator with varying cell temperatures, and the results showed that cell temperature has a significant effect on the PV parameters and controls the quality and performance of the solar cell.

The current literature has also shown that there are many advantages and disadvantages of using each cooling method. The advantages and disadvantages of using the standardized cooling methods of air, PCM, and water are represented in Figure 2.

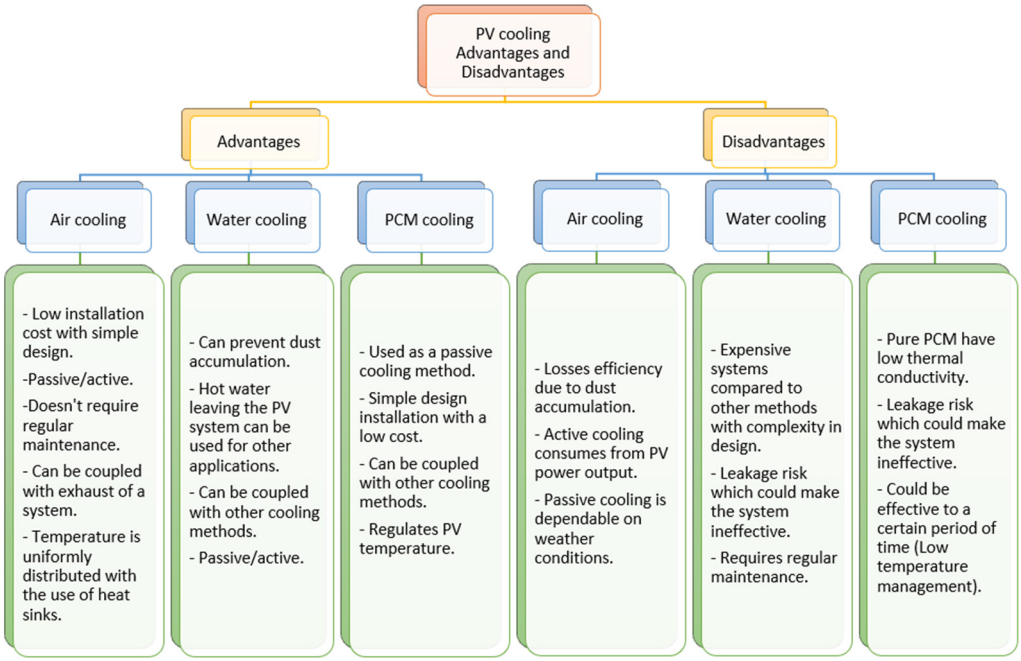


Figure 2. Advantages and disadvantages of cooling methods.

2.4. Governing Equations

Every PV panel has a length L, a width W, and a thickness t. To calculate the total area of a PV panel, then,

$$A = L \times W \tag{1}$$

where

A: Area of the P panel (m^2).
L: Length of the PV panel (m).
W: Width of the PV panel (m).

However, the effective area of the PV panel is the area which yields power. This area can be calculated as:

$$A_{eff} = A_{cell} \times nb_{cell} \tag{2}$$

where

A_{eff}: Effective area of the PV panel (m^2).
A_{cell}: Area of one cell (m^2).
nb_{cell}: Number of cells in a PV panel.

The power received from the sun is:

$$Q_{solar} = G \times A_{eff} \times \alpha \times \tau \quad (3)$$

where

Q_{solar}: Solar energy falling perpendicularly on the frontal surface of the PV panel as an input power (W).

G: Solar radiation intensity incident on the panel in $(\mathrm{W/m^2})$.

α: Glass absorptivity.

τ: Glass transmissivity.

The power output of the PV panel is calculated by:

$$P_{elect} = V \times I \quad (4)$$

where

P_{elect}: Electric power output of the PV panel (W).

V: Output voltage (V)

I: Output current (A).

The output voltage and currents could be measured by mustimeters, where the voltage is measured in parallel and the current in series.

The electric efficiency of a PV panel is measured using:

$$\eta_{elect} = \frac{P_{elect}}{P_{solar}} \times 100 \quad (5)$$

where

η_{elect}: Electric efficiency $(\%)$.

P_{elect}: Output electric power (W).

P_{solar}: Input electric power (W).

The solar incident angle is the angle between the perpendicular and the incoming light from the sun. It is quantified by:

$$AOI = \cos^{-1}\left[\cos(\Theta_z)\cos(\theta_T) + \sin(\Theta_z)\sin(\theta_T)\cos(\theta_A - \theta_{A_{array}})\right] \quad (6)$$

where

Θ_z: The solar zenith angle.

θ_T: The tilt angle of the array.

θ_A: The solar azimuth angle.

$\theta_{A_{array}}$: The azimuth angle of the array.

The installation angle of a PV panel is the same as the tilt angle. It is the angle between the horizontal surface and the PV panel. It is quantified as:

For the northern hemisphere:

$$\alpha = 90^\circ - (\phi - \delta) \quad (7)$$

For the southern hemisphere:

$$\alpha = 90^\circ + (\phi - \delta) \quad (8)$$

where

ϕ: The latitude.

δ: The angle of declination.

3. PV Cooling Methods

Efficiency improvement of PV panels depends mainly on mitigating panel temperature. Figure 3 shows the three main cooling techniques in addition to other not-well-known

and new techniques. The water cooling technique involves an earth water heat exchanger, solar water disinfection, a heat pipe system and an automotive radiator system. These methods are classified as either active or passive methods. The phase-change material (PCM) cooling technique is divided into organic PCM and non-organic PCM, while the air cooling method is divided into the installation of heat sinks, jet impingements, air duct or cavity air flow systems to the PV panel. These air cooling methods are classified as forced or free convection systems. Finally, non-categorized cooling methods are divided into the thermoelectric cooling method, the coating method and nanofluids. These methods are either new or not well known compared to the other cooling techniques.

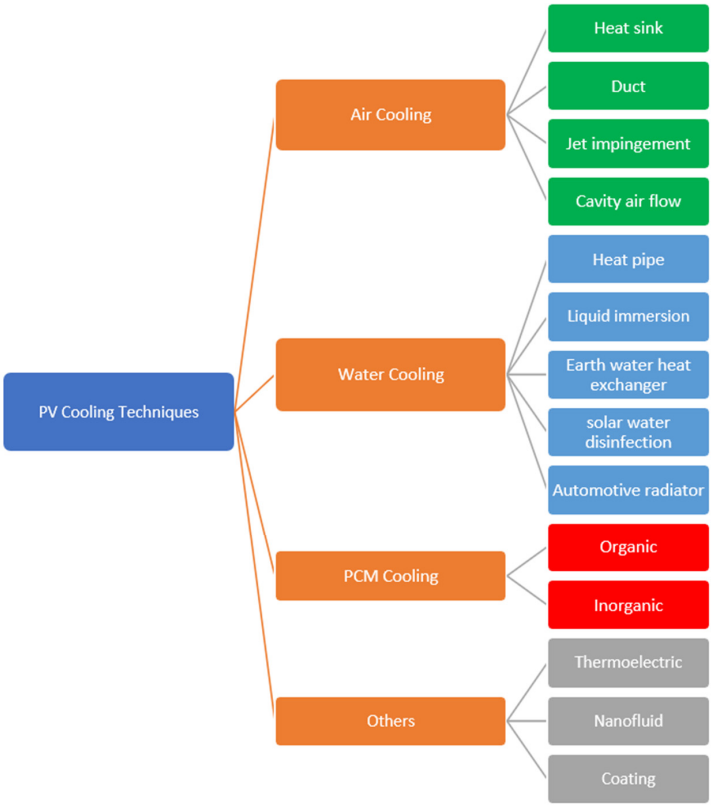


Figure 3. Classification of cooling techniques.

3.1. Air Cooling Methods

The air cooling method for PV refers to the technique of dissipating heat from PV modules by circulating air around them. It can be implemented in free or forced convection, using heat sinks, fans, or blowers to increase airflow. As shown in Figure 4, natural convection occurs by the means of circulation and heat exchange between hot and cold fluids, this circulation is caused by the buoyancy effect. When the PV panel becomes hot, it warms up the layer of air surrounding it, thus the temperature of air increases, and the density increases accordingly. Consequently, hot air rises, causing a movement called a natural convection current.

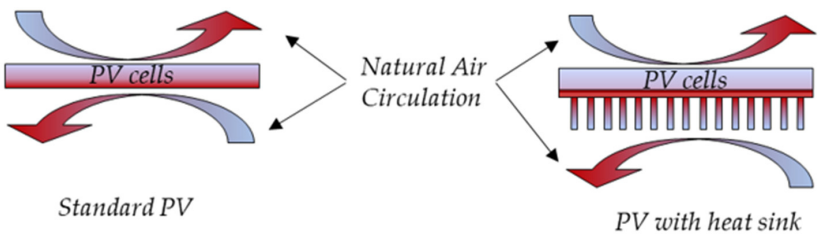


Figure 4. PV panel under free convection with or without a heat sink.

Forced convection is considered one of the most effective heat transfer mechanisms. It is characterized by using external sources such as fans, pumps, and suction devices to aid fluid transportation.

Air cooling is relatively simple and cost-effective, making it a popular choice for cooling PV systems. However, its effectiveness depends on various factors such as ambient temperature, humidity, and wind speed. Heat sinks can be used in conjunction with air cooling to further improve heat dissipation and maintain a stable operating temperature for the PV modules.

Below, we present a summary table that outlines various cooling techniques with both free and forced convection methods for photovoltaic panel cooling.

Table 2 summarizes various cooling methods applied to photovoltaic panels to enhance their efficiency under different convection conditions. The studies cover a spectrum of techniques, including forced convection with ducts and fans, free convection using multi-level fin heat sinks, and hybrid approaches combining free and forced convection with phase-change materials. Results indicate notable improvements in efficiency, ranging from 2.1% to 21.68%, with specific configurations achieving enhanced performance in different climates. Additionally, studies explore novel strategies such as curved eave and vortex generators, graphite-infused PCM, and heat spreaders with cotton wicks. Overall, the studies explore a range of cooling methods and their impacts on PV panel performance, contributing valuable insights to the field of renewable energy.

Moreover, the numerical studies in Table 2 have shown more novel approaches in the designs of the cooling methods used in cooling the PV panel. Numerical investigations shown a temperature reduction ranging between 5.89 °C and 27 °C while mainly focusing the studies on using free convection. However, experimental investigations were combining both free and forced convection and comparing their results. Air cooling was found to be effective in significant solar radiation climates, where the temperature of the air is lower than the temperature of the PV's operating temperature.

Table 2. Free and forced convection cooling methods.

Convection Method	Cooling Method	Test Methodology	Results	Climate	Author
Forced convection	PV 1: Lower duct with blower. PV 2: Duct with DC fans	Experimental and Numerical	Enhanced efficiency by 2.1% and 1.34% using fans and blower, respectively	Benha, Egypt	Hussein et al. [30]
Free Convection	Truncated multi-level fin heat sink	Numerical	Recorded a 6.13% temperature decrease and a 2.87% increase in output power	-	Ahmad et al. [31]
Free convection and forced convection	PV 1: Heat sink under free convection. PV 2: Duct under forced convection. PV 3: Fins in a duct under forced convection. PV 4: PCM (White petroleum jelly with a melting point of 37 °C)	Experimental	Improvements in efficiency by 0%, 33%, 53%, and 72% for PCM, heat sink under free convection, duct under free convection, and duct under forced convection, respectively	Kumasi, Ghana	Abdallah et al. [32]
Free convection	PV 1: Free convection using through-holes in the PV panel. PV 2: Active water spraying on the surface. PV 3: Passive and active cooling using through-holes in the PV and water spraying on surface	Numerical	Hybrid cooling resulted in an average reduction of 17.24 °C	-	Pomares-Hernández et al. [33]
Forced convection	Curved eave and vortex generators	Numerical	Achieved a 5.89 °C temperature reduction	-	Wang et al. [34]
Forced convection	PVT system under forced convection by DC fans	Experimental	Electric efficiency between 12% and 12.4% with 0.05 m channel depth and 0.018 kg/s to 0.06 kg/s air mass flow rate	Tehran, Iran	Kasaeian et al. [35]
Free convection	PV 1: 30 mm graphite-infused PCM (paraffin wax with a melting point of 40 °C). PV 2: Finned heat sink. PV 3: Finned heat sink with graphite-infused PCM	Experimental and Numerical	Finned heat sink with graphite-infused PCM demonstrated an overall efficiency increase of 12.97%	New Zealand (Laboratory)	Atkin et al. [36]

Table 2. *Cont.*

Convection Method	Cooling Method	Test Methodology	Results	Climate	Author
Free convection	Heat spreader with cotton wicks	Experimental	Recorded a 12% decrease in temperature and a 14% increase in electric output	Tamil Nadu, India	Chandrasekar et al. [37]
Free convection	Twisted baffle at the rear surface of the PV	Numerical	Efficiency increases by 1.21% and 3.36% for solar radiation of 200 W/m^2 and 1000 W/m^2, respectively	-	Benzarti et al. [38]
Free convection	PV 1: L-profile aluminum fins with parallel configuration. PV 2: L-profile aluminum fins randomly positioned	Experimental	Electric efficiency increased by 2% for L-profile aluminum fins with random distribution	Split, Croatia	Grubišić-Čabo et al. [39]
Forced convection	PV/T system with rectangular finned plate	Experimental	Recorded a maximum efficiency of 13.75% for 4 fins under solar radiation of 700 W/m^2 and a mass flow rate of 0.14 kg/s	Malaysia	Mojumder et al. [40]
Free convection	Cooling tower with PV module	Numerical	Averaged 6.83% increase in annual efficiency of the PV	-	Abdelsalam et al. [41]
Forced convection	PV 1: Air from above and water from below. PV 2: Air from above and below. PV 3: Air from above. PV 4: Air from below. PV 5: Water from below	Numerical	Water below the PV panel decreased the temperature by 21 °C	Sakaka Al-Jouf, KSA	Soliman [42]
Free convection	Effect of using the racking structure of the PV panel system as a passive heat sink for cooling	Experimental and Numerical	Achieved a 3% increase in electric efficiency with a 6.3 °C PV temperature reduction	Dammam, Saudi Arabia	El-Amri et al. [43]
Free convection	Investigated the use of different dimensions of a finned plate in cooling the PV panel	Experimental	Utilizing a 7 cm by 20 cm staggered fin array resulted in the best performance with an energy efficiency of 11.55%	Elazig, Turkey	Bayrak et al. [44]
Free convection	Studied the effect of dust accumulation density on the convective heat transfer coefficient for a large-scale PV panel array	Experimental	Increased convective heat transfer coefficient by 4.13% compared to a clean PV module	Zhongwei, Ningxia province in China	Hu et al. [45]
Free and forced convection	PV 1: PV-duct under free convection. PV 2: PV-duct under forced convection. PV 3: PV-duct under forced convection with L-shaped barrier	Experimental	The highest electric efficiency of 21.68% was recorded by the PV panel under forced convection with an L-shaped barrier in its duct	India	Kumar et al. [46]
Free convection	PV–heat sink system with different fin dimensions	Numerical	The initial heat sink model was able to cool the PV panel by 27 °C	Dubai, UAE	Mankani et al. [47]
Free convection	PV module with porous material. This study was performed on three porous fins, a porous layer, and five porous fins	Numerical	Increase of 6.73%, 8.34%, and 9.19% in efficiency for the three porous fins, porous layer, and five porous fins configurations, respectively	-	Kirwan et al. [48]
Forced convection	PV-compressed air module	Numerical	This method improved the output power of the PV panel and as a result, improved its efficiency	-	Li et al. [49]

3.2. Water Cooling Methods

PV water cooling methods are a set of techniques that involve the use of water or other fluids to absorb and dissipate heat from PV panels, with the goal of improving their electrical performance and prolonging their lifespan. These methods can be implemented through passive or active means and may involve the use of heat sinks, heat exchangers, direct water immersion, or other related approaches. The effectiveness of PV water cooling methods depends on various factors, such as water flow rate, temperature, and quality, as well as the design and construction of the cooling system. Table 3 represents the different cooling techniques that are either passive or active.

Table 3. Classification of passive and active water-based cooling techniques.

Passive Cooling Techniques	Active Cooling Techniques
Liquid immersion Heat pipe	Earth water heat exchanger Solar water disinfection Automotive radiator

Passive cooling techniques for cooling PV systems refer to natural methods used for reducing the temperature of PV modules without the use of mechanical or electrical devices. They rely on convection, radiation, and evaporation to dissipate heat and improve the performance and lifespan of PV modules.

Tina et al. [50] have increased the electrical efficiency by approximately 10% after experimentally submerging a PV panel inside water in a study of enhancing PV temperature. Figure 5 represents the submerged PV inside a vessel containing water.

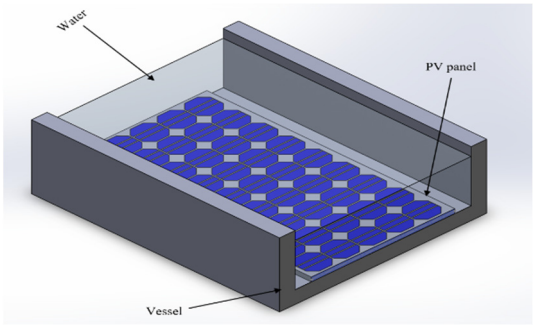


Figure 5. PV panel immersed in water [50].

On the other hand, active water cooling for PV required a mechanical or electrical devices to actively reduce the temperature of PV modules. This may include circulating water or other fluids through a heat exchanger. They are useful in hot climates or high-power output systems and provide greater cooling efficiency and control over operating temperature than passive cooling methods.

Irwan et al. [51], carried an indoor experiment in order to investigate the effect of water flowing at the surface in cooling the PV panel. Results showed that a decrease in PV temperature by 5–23 °C increases the output power of the PV panel by 9–22%.

On the other hand, Moradgholi et al. [52] experimentally investigated the effect of heat pipes in cooling PV panels, and the module used in his experimental study is represented in Figure 6. Results showed an increase of 5.67% in power when using methanol as a working fluid in spring and an increase of 7.7% in power when using acetone as a working fluid in summer.

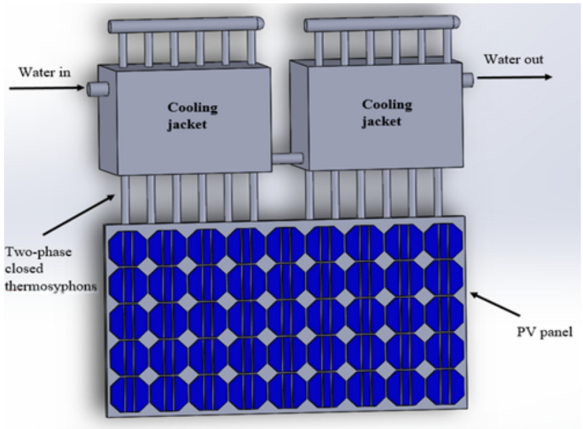


Figure 6. Heat pipes module [52].

Moreover, Sandeep Koundinya et al. [53] investigated experimentally and by simulation the effect of a finned heat pipe with water as the working fluid in cooling photovoltaic panels. Results showed a total decrease of 13.8 K in PV panel temperature and good agreement was found between experimental and computational studies.

Below, a summary table is presented for several studies about cooling PV modules with passive and active cooling techniques.

Table 4 presents a wide array of outcomes across various cooling methods for photovoltaic panels. Passive approaches, like water-saturated microencapsulated phase-change materials (MEPCM) and immersion in dielectric liquids, effectively reduce temperatures, leading to improved electric efficiency. Passive cooling techniques exhibit diverse results, with efficiency enhancements ranging from 2.7% to 12.4% and a temperature reduction of up to 13.8 K. Active cooling methods, such as spraying water and flowing water on the PV surface, consistently boost power generation and efficiency, demonstrating improvements from 8% to 9% to a significant 24 K temperature decrease. Innovative methods like floating PV on water surfaces and geothermal cooling systems show efficiency increases of 2.7% and up to 13.8%, respectively. The choice between passive and active cooling depends on factors like climate, available resources, and desired efficiency levels. These findings collectively contribute to advancing PV panel cooling, facilitating more efficient and sustainable solar power generation.

Table 4. Summary of several studies on water-based cooling techniques for PVs.

Cooling Method	Cooling Classification	Test Methodology	Key Outcomes	Climate	Author
Water-saturated microencapsulated phase-change material (MEPCM)	Passive	Numerical	A layer of PCM of 5 cm thickness with a melting temperature of 30 °C gave the best performance in enhancing the electric efficiency.	-	Ho et al. [54]
Liquid immersion of solar cells in 4 different dielectric liquids.	Passive	Numerical	Immersing the solar cells in the dielectric liquids maintained a low temperature in the solar cells.	-	Liu et al. [55]
Spraying water on frontal and rear surfaces.	Passive	Experimental	Increase in power and efficiency by 16.3% and 14.1%, respectively.	Croatia	Nizetic et al. [56]
Finned heat pipe system with water as a working fluid.	Passive	Experimental and Numerical	A total decrease of 13.8 K in PV panel temperature and good agreement was found between experimental and computational studies.	India	Koundinya et al. [53]
PV/T system with water and ethylene glycol as working fluids.	Passive	Experimental and Numerical	Water was found to be a better coolant than ethylene glycol with an overall efficiency enhancement by 25%.		Joy et al. [57]
Spraying water on surface.	Active	Experimental and Numerical	Cooling system have good performance in hot and dusty regions.	Egypt	Moharram et al. [58]
Flat-plate PV/T system with and without glass cover.	Active	Numerical	Empirical correlations were performed and conclusions were conducted.	-	Bajestan et al. [59]
Flowing water on PV surface.	Active	Experimental	Increase in power by 8–9%	Laboratory	Krauter [60]
Heat pipe.	Active	Numerical	As the convective heat, transfer coefficient increases the solar cells temperatures decreases when operating at low flow rates and at high optical concentration ratios.	-	Sabry [61]
Spraying water on the PV surface.	Active	Experimental	Increase of 2.7% in electrical efficiency and 21 W in power.	Alexandria, Egypt	Elnozahy et al. [62]
Flowing water on the surface.	Active	Experimental	Increase in the power generated and in total efficiency.	Iran	Kordzadeh et al. [63]
Water system with air blowing to the back of the PV.	Active	Numerical	Yearly improvement of 5% in efficiency.	-	Arcuri et al. [64]
Earth water heat exchanger.	Active	Numerical	Increasing the length of the feed pipe to 60 m would decrease PV temperature by 23 °C.	Pilani, Rajhasthan, India	Jakhar et al. [65]

Table 4. *Cont.*

Cooling Method	Cooling Classification	Test Methodology	Key Outcomes	Climate	Author
Concentrated PV/T system.	Active	Numerical	Empirical correlations were performed and conclusions were conducted.	-	Mittelman et al. [66]
Automotive radiator.	Active	Experimental and Numerical	Theoretical heat rejection by 91% and experimental efficiency increased by 4.46%.	Kuala Lumpur, Malaysia	Chong et al. [67]
Solar desalination combined with an intermittent solar-operated cooling unit.	Active	Experimental	A 13.75% energy efficiency for the system.	Cairo, Egypt	Ibrahim et al. [68]
PV/T system laminated with polymer matrix composite with water as a coolant.	Active	Experimental and Numerical	The maximum efficiency recorded was 20.8% with a 53.5% thermal efficiency.	-	Korkut et al. [69]
PV 1: single-pass ducts. PV 2: multi-pass ducts. PV 3: tube-type heat absorber. Water is used as a fluid.	Active	Numerical	Cell temperature achieved a maximum of 38.310 °C.	Islamabad, Pakistan	Sattar et al. [70]
Flowing water on the PV surface.	Active	Experimental (laboratory and real-life conditions)	The system showed a temperature decrease of 24 K with a power generation increase of 10% with a return on investment of less than 10 years.	Krakow, Poland	Sornek et al. [71]
Floating PV on the water surface.	Passive	Experimental	An efficiency increase of 2.7% was recorded with a temperature decrease of 2.7 °C	Cagliari, Italy	Majumder et al. [72]
A new innovative cooling box acting as a thermal collector.	Active	Numerical	Electric efficiency of 17.79% and a thermal efficiency of 76.13% when the system was studied with a mass flow rate of 0.014 kg/s and an inlet water temperature of 15 °C.	-	Yildirim et al. [73]
Water flows on the surface of the PV panel.	Passive	Experimental	An increase in exergy efficiency from 2.91% to 12.76%.	Sisattanark district, Vientiane Capital, Laos	Chanphavong et al. [74]
Comparison between water flowing on the surface of the PV panel and wet grass cooling.	Passive	Experimental	Running water on the upper surface of the PV helps in cooling it and increasing its efficiency.	Gwalior, India	Panda et al. [75]
Comparison between conventional PV panels, concentrated PV systems, and water-cooled concentrated PV systems.	Active	Experimental and Numerical	Significant increase in the efficiency and power output of the water-cooled CPV system to 17% and 23%, respectively. The overall output power of the water-cooled CPV was 24.4%.	Duhok, North of Iraq	Zubeer et al. [76]
A geothermal cooling system containing a mixture of water and ethylene glycol.	Active	Experimental	An increase in electric efficiency up to 13.8% using a constant coolant flow rate of 1.8 L/min.	Alcalá de Henares, Madrid, Spain	Lopez-Pascual et al. [77]
Radiative cooling module.	Active	Experimental	Increase in efficiency by 1.21% and 0.96% in summer and autumn, respectively, for the system without cold storage. For the system with a cold storage, the efficiency increased by 1.69% and 1.51% in summer and autumn, respectively.	China	Li et al. [78]
Porous media with water as a cooling fluid.	Active	Experimental and Numerical	Decrease by 35.7% of PV's surface temperature and increase by 9.4% in the output power under a volume flow rate of 3 L/m with a porosity of 0.35.	Jordan	Masalha et al. [79]
Geothermal heat exchanger with water and ethylene glycol as cooling fluids.	Active	Experimental and Numerical	Increase in PV's electric power generation by 9.8%.	Turkey	Jafari et al. [80]
Water cooling system and phase-change material (PCM) module with OM35 as a PCM with a melting point of 35 °C.	Passive	Experimental	Increase in the electric efficiency by 12.4% compared to the other configurations.	Chennai, India	Sudhakar et al. [81]

The use of evaporative cooling could be more beneficial than vapor compression at the level of the cost. However, the system is not reliable or needs more design work [82].

Moreover, it was noticed in the water-cooled methods that the experimental studies mentioned in Table 4 were greater than the numerical studies and the climates the water cooling methods were studied in are hot such as India, Egypt, and Saudi Arabia.

3.3. PCM Cooling Methods

Phase-change materials (PCMs) are substances used in cooling systems for photovoltaic modules to absorb and store heat from the panels during peak sunlight hours. PCMs have a high latent heat of fusion, which means they can absorb large amounts of heat without a significant increase in temperature. PCMs can be integrated into PV panels, or used in a separate thermal management system to enhance the overall efficiency and lifetime of the PV system.

In a typical PV–PCM hybrid system, illustrated in Figure 7, the PCM functions as a heat sink that absorbs excess heat from the PV panel, thereby reducing its temperature. During the peak sun hours, the temperature of the PV panel exceeds the melting temperature of the PCM. As a result, the PCM absorbs excess heat from the PV panel and maintain a stable operating temperature for the PV system until it completely melts, transitioning from solid to liquid phase. During the low sunlight period, as the ambient temperature decreases and the temperature of the PV panel drops below the melting point of the PCM, the PCM releases excess heat and solidifies again.

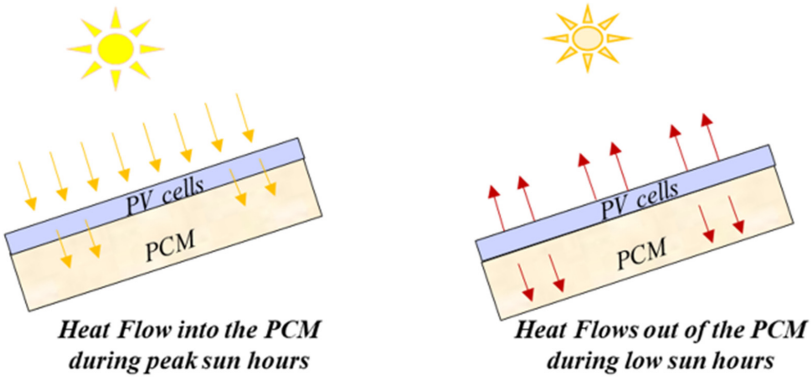


Figure 7. Typical PV–PCM system [83].

Table 5 summarizes the various cooling techniques using PCM with different combinations and materials.

Table 5. Summary of PV cooling techniques based on PCMs.

PCM Used	PCM Melting Point	Cooling Method	Test Methodology	Key Outcomes	Climate	Author
Pure PCM: white petroleum jelly. Combined PCM: white petroleum jelly + graphite + copper	36–60	Pure and combined PCM	Exp.	Efficiency increased by an average of 3% when using pure PCM and by an average of 5.8% when using combined PCM.	Bekaa Valley, Lebanon.	Hachem et al. [84]
-	0–50	PV panel containing an integrated layer of PCM	Num.	Efficiency exceeds 6% in some regions.	-	Smith et al. [85]
RT25	25	Impure PCM layer integrated into the PV panel	Num.	Maintain panel operating temperature under 40 °C for 80 min under solar radiation of 1000 W/m^2.	-	Biwole et al. [86]
Salt hydrate, $CaCl_2{\cdot}6H_2O$ and eutectic of capric acid–palmitic acid	$CaCl_2{\cdot}6H_2O$: 29.8 and eutectic of capric acid-palmitic acid: 22.5	PCM layer with aluminum alloy fins integrated into the PV panel	Exp.	$CaCl_2{\cdot}6H_2O$ showed an increased power output of 3% compared to capric-palmitic acid in Pakistan. The two PCMs showed better results in Vehari, Pakistan than in Dublin, Ireland with a total of 13% in power saving.	Dublin, Ireland and Vehari, Pakistan	Hasan et al. [87]
Paraffin wax	37.5–42.5	PV–PCM system	Exp.	Average maximum efficiency and power were increased by 1.63% and 1.35 W, respectively.	Laboratory	Xu et al. [88]
Rubitherm 28 HC and Rubitherm 35 HC	Rubitherm 28 HC: 27–29 and Rubitherm 35 HC: 34–36	PV–PCM system	Num.	Increase by 10% in peak power and 3.5% in energy produced throughout the whole year round.	-	Aneli et al. [89]

Table 5. *Cont.*

PCM Used	PCM Melting Point	Cooling Method	Test Methodology	Key Outcomes	Climate	Author
Docosane paraffin wax	42	PV–PCM system	Num. and Exp.	An increase of 1.05% in efficiency and a 34% increase in life span.	Doha, Qatar.	Amalu et al. [90]
RT35HC	36	PV–PCM system	Num.	Temperature reduction by 24.9 °C and an increase of 11.02% in electric output.	-	Zhao et al. [91]
RT35	35	PV–PCM system	Num.	Total increase of 5% in productivity.	-	Kant et al. [92]
-	24.85	PV–PCM system	Num. and Exp.	Increase of 1–1.5% in electric efficiency.	Song-do, Incheon, South Korea	Park et al. [93]
RT28HC	28	PV–PCM system	Exp. and Num.	Increase in power by 9.2% experimentally and 4.3–8.7% numerically.	Ljubljana, Slovenia	Stropnik et al. [94]
-	23	BIPV–PCM	Exp. and Num.	Maximum electric and thermal efficiencies recorded were 10% and 12%, respectively.	Lisbon, Portugal	Aelenei et al. [95]
Paraffin wax	34.9–42	PV–PCM system and PV–PCM thermal system	Exp.	An electric output increase of 5.18% in the PV–PCM system and 30.4% electric sum was recorded in the PV–PCM-T system.	Shanghai, China	Li et al. [96]
RT42	38–43	PV–PCM system	Exp.	Annual enhancement of 5.9% in electric yield in hot climate.	Al Ain, United Arab Emirates	Hasan et al. [97]
RT27 & RT31	RT27: 25–28 and RT31: 27–33	PV–PCM systems	Exp.	Enhancement in energy by 4.19% and 4.24% when using RT27 and RT31, respectively.	Chania, Greece	Savvakis et al. [98]
Eutectic of capricpalmitic acid and calcium chloride hexahydrate and RT20 and RT25 and RT35	Eutectic of capricpalmitic acid: 22.5 and calcium chloride hexahydrate: 29.8 and RT20: 25.73 and RT25 26.6 and RT35: 29–36	PV-T-nano-PCM system	Num.	Increase in electric efficiency by 6.9% and 22% in winter and summer weather, respectively.	Dhahran, Saudi Arabia	Abdelrazik et al. [99]
Paraffin A44	44	PVT–PCM system	Exp. and Num.	Electric performance increased by 7.2% numerically and 7.6% experimentally.	Kuala Lumpur, Malaysia	Fayaz et al. [100]
Lauric acid	44–46	PVT–PCM system	Exp.	PVT–PCM system increased the electric efficiency of the PV by 1.2% under a volume flow rate of 4 LPM.	Kuala Lumpur, Malaysia	Hossain et al. [101]
Paraffin wax	46–48	PVT–PCM system with pure water and ethylene glycol as working fluids	Exp.	Energy loss percentage was decreased by 9.28%, 23.33%, and 48.58% for the PVT/water, PVT/ethylene glycol (50%), and PVT/ethylene glycol (100%).	Mashhad, Iran	Kazemian et al. [102]
RT25	26	PV–PCM and PV–PCM–fins systems	Num.	Increase of 2.5% and 3.5% in electric efficiency in fair and sunny weather, respectively.	Different weather conditions	Metwally et al. [103]
RT58, RT42, and C22-C40	RT58: 58 and RT42: 42	PV–PCM–heat sink system	Num.	Temperature drop of 18.3 °K, 21.2 °K, and 26.1 °K when using C22-C40, RT58, and RT42, respectively.	Oujda, Morocco	Bria et al. [104]
-	273.15 K	PV–PCM matrix absorber system	Num.	Analytical and numerical results were in agreement.	-	Hassabou et al. [105]
-	-	PV/T system with nano-enhanced MXene-PCM and R407C working fluid	Num.	Power output increased by 535 KWh/year and electric efficiency increased by 3.01%.	Derby, United Kingdom	Cui et al. [106]
-	-	PV–PCM system with a heat sink with convex/concave dimples	Num.	PV cell temperature decreased by 7.14%, 4.65%, and 2.22% when studying the PV cells at inclinations of 90°, 60°, and 30°, respectively.	-	Soliman et al. [107]
Paraffin wax	38–43	PV–PCM system	Exp.	Efficiency was improved by 14.4% when using a PCM thickness of 3 cm and tilting the PV at an angle of 30°.	Qena, Egypt	Maghrabie et al. [108]
Paraffin wax and vaseline	Paraffin wax: 45 Vaseline: 25	Water-cooled PVT system with PCM	Exp.	Increase in electric and thermal efficiencies up to 13.7% and 39%, respectively.	Indoor (Simulating Iraq's weather)	Chaichan et al. [109]
RT28HC	25–29	PV–PCM system	Exp.	Enhancement by 2.5% in the power output.	Mediterranean climate	Nizetic et al. [110]

Table 5 presents a comprehensive overview of different phase-change materials utilized in conjunction with photovoltaic (PV) panels. Each entry includes details on the specific PCM used, its melting point, cooling method, test methodology, key outcomes, climate conditions, and the contributing authors. Noteworthy PCMs like white petroleum jelly, paraffin wax, and specialized formulations such as Rubitherm 28 HC and Rubitherm 35 HC are explored across various cooling systems. Passive cooling methods incorporating PCMs exhibit efficiency gains ranging from 1.05% to 12.4%. On the other hand, active systems like PV–PCM configurations and PV/T systems consistently showcase improvements in electric efficiency and power output, reaching up to 24%. These findings underscore the impact of factors such as climate, location, and PCM composition on the effectiveness of these cooling techniques. Overall, Researchers have identified optimal PCM parameters, thicknesses, and integration methods, contributing to the advancement of efficient photovoltaic cooling strategies.

Moreover, cooling by PCM is shown to be used in hot climates where solar irradiation is considered to be high such as in Egypt, Iran, Saudi Arabia, and United Arab Emirates.

3.4. Other Cooling Methods

There are various cooling methods for photovoltaic systems other than air, water, and phase-change materials. One of these methods is using encapsulated PCM, which improves the PCM's expansion property during melting. Another method is thermoelectric cooling, which uses the Peltier effect to create a temperature difference and transfer heat away from the PV module. Additionally, researchers have explored the use of nanofluids, which are liquids containing nanoparticles that can improve the thermal properties of the cooling fluid. Other methods include using refrigeration systems or hybrid systems that combine multiple cooling methods. Each of these cooling methods has its own advantages and disadvantages and can be suitable for different types of PV systems and operating conditions.

Saleh et al. [111] numerically studied the effect of nanofluid and water in cooling photovoltaic–thermal (PVT) collectors. Results showed that the use of 1% volumetric fraction of nanofluids increases the thermal efficiency up to 19.5% and the electric efficiency up to 55.45%.

Ghadiri et al. [112], experimentally studied the effect of cooling a PVT system by ferrofluids shown in Figure 8. Different fluids were used at a constant and an alternating magnetic field in order to discuss the effect of magnetic field on ferrofluids. Results showed an increase of 45% in the overall efficiency when using ferrofluid and a total increase of 50% in the overall efficiency when using an alternating magnetic field of 50 Hz frequency. Also, a total of 48 W of exergy was increased after using ferrofluid.

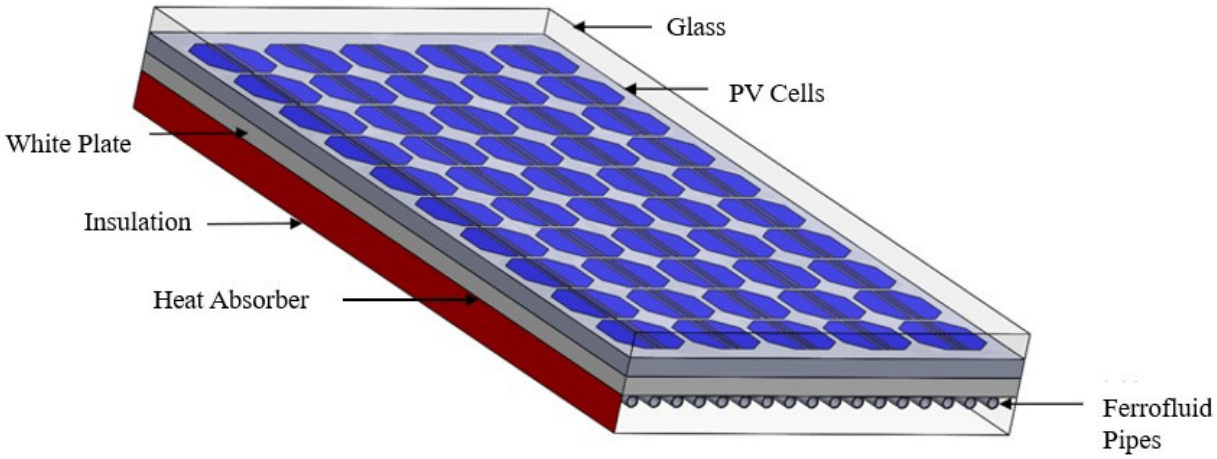


Figure 8. Ferrofluids cooling system [112].

Figure 9 illustrates the use of thermoelectric water-nanofluid cooling and thermoelectric finned heat sink cooling in PV/T and PV systems respectively [113].

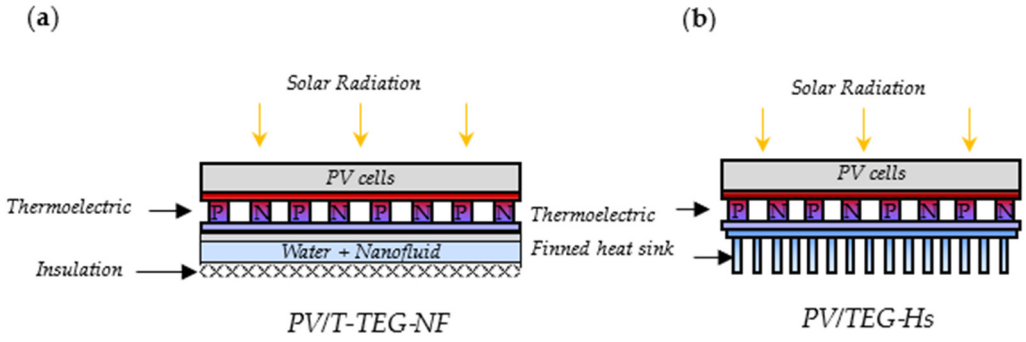


Figure 9. PV with thermoelectric plate and: (**a**) photovoltaic/thermal-thermoelectric water-nanofluid (PV/T-TEG-NF) cooling system and (**b**) photovoltaic-thermoelectric finned heat sink (PV/TEG-Hs) cooling system [113].

Table 6 shows the recent studies performed on cooling the PV panel using different methods.

Table 6. Different cooling techniques.

Cooling Method	Test Methodology	Key Outcomes	Climate	Author
Microencapsulated PCM heat sink with a thermoelectric generator	Experimental	Increase in efficiency by 2% in the intermediate season and by 2.5% in the summer.	Republic of Korea	Kang et al. [114]
PCM-integrated PV system with fins and nanofluid (CPV/T/NF/FPCM)	Experimental	Electric efficiency was improved up to 17.02% while thermal efficiency was improved up to 61.25%.	Tehran, Iran	Kouravand et al. [115]
Photovoltaic thermal collector with a nano-PCM and micro-fin tube nanofluid system	Experimental	The micro-fins, nanofluids, and nano-PCM PV had a thermal efficiency of 77.5% with an increase in electric power of 4.01 W.	Indoor (Solar Simulator)	Bassam et al. [116]
Micro-fin tube counterclockwise twisted tape nanofluid and nano-PCM	Experimental	Increase of 44.5% in electric power.	Indoor (Solar Simulator)	Bassam et al. [117]
PV/nano-enhanced PCM heat sink system	Experimental	The GNP-CuO 3% mixture has enhanced the thermal conductivity by 91.81%, reduced temperature by 6.6 °C, and enhanced the electricity output by 3%.	Iran	Moein-Jahromi et al. [118]
PCM, thermoelectric cooling, and installing fins made of aluminum in cooling the PV panel	Experimental	The PV panel with aluminum fins had the highest power generation enhancement of 47.88 watts.	Elazig, Turkey	Bayrak et al. [119]
PV/T with spectrum-splitting module	Numerical	Conversion efficiency exceeded 43%.	-	Xu et al. [120]

The outcomes presented in Table 6 highlight the diverse and innovative cooling methods for photovoltaic panels. The utilization of a microencapsulated phase-change material combined with a heat sink, and a thermoelectric generator, demonstrated a 2% efficiency increase in the intermediate season and 2.5% in summer. The integration of PCM with fins and nanofluid (CPV/T/NF/FPCM) showed significant improvements, achieving an electric efficiency of 17.02% and a thermal efficiency of 61.25%. Indoor experiments involving a photovoltaic thermal collector with Nano-PCM and micro-fin tube nanofluid revealed a remarkable thermal efficiency of 77.5% and a 4.01 W increase in electric power. Another noteworthy system, incorporating a micro-fin tube counter clockwise twisted tape nanofluid and nano-PCM, demonstrated a substantial 44.5% increase in electric power. The PV/nano-enhanced PCM heat sink system displayed enhancements, including a 91.81% increase in thermal conductivity, a 6.6 °C temperature reduction, and a 3% improvement in electricity output. Further experiments, incorporating PCM, thermoelectric cooling, and aluminum fins, yielded the highest power generation enhancement of 47.88 Watts. Additionally, a numerical simulation of a PV/T system with a spectrum-splitting module revealed an impressive conversion efficiency exceeding 43%. These advancements hold promise for improving the energy efficiency and sustainability of photovoltaic panels and increasing the adoption of renewable energy sources.

4. Discussion and Analysis

The literature has provided numerous methods for cooling the PV panel and increasing its efficiency, resulting in methods with more effectiveness over others. With different methods of cooling, different ranges of efficiency arise that were obtained and illustrated in Figure 10.

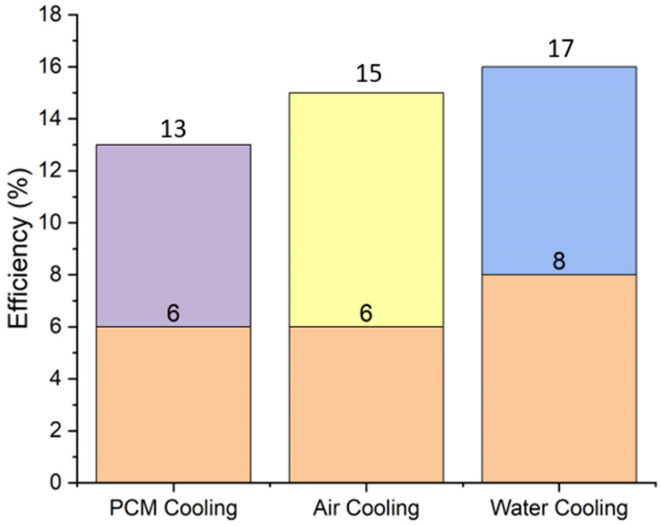


Figure 10. Efficiencies of cooling methods.

According to the literature, efficiency ranged between 6% and 13% when PCM was used as a cooling technique. This method had both advantages and disadvantages. Through all the previous studies, the main problem that faced the researchers is the low thermal conductivity of PCM, and the change in volume when PCM melts, which in turn leads to poor temperature management. Researchers tried to solve these problems through several ways such as mixing PCM with graphite and developing a shape-stabilized PCM. In contrast, the advantages of this method were the simplicity of the cooling system, the low cost, and the long lifetime. With the need for only a tank filled with PCM attached to the back of the PV panel, the price of this system was low, and with the absence of electrical instruments, there will be no need for maintenance.

The air cooling techniques literature revealed a range of efficiency between 6% and 15%, and several methods were tested experimentally and numerically based on natural convection and forced convection. Several systems were tested by scientists such as finned plates, fans and air ducts, finned plates combined with fans and air ducts, and jet impingements. Others tried to combine the effects of the latent heat storage of PCM along with finned plates under natural and forced convection. The simplest method and most effective was using finned plates under forced or natural convection. Under forced convection, a better efficiency was recorded but a higher cost compared with the use of finned plates under natural convection. Therefore, there is no method better than another in general; but in specific conditions, optimization between efficiency and cost can be achieved. In a windy location, a finned free convective system will give great efficiency with low cost, while in a non-windy location, a finned forced convective system would cost a little bit more but will give a higher efficiency.

The highest efficiency was recorded when water cooling systems were tested. Different techniques were taken into consideration, spraying water over the surface of the panel, immersion of the panel in water, using water as a circulation fluid in heat pipes attached to the back of the PV, etc. Efficiency with water systems ranged in the literature between 8% and 17%, but designing systems to deal with water had a high cost because of the need for pumps, pipes, fittings, etc. In addition, when taking the location of the project into consideration, a water cooling system was the best technique in dusty or sandy places, where high efficiency could be maintained by removing dust from the front surface of the panel which would otherwise reduce the amount of irradiation received.

5. Economic Study

After cooling the PV panels, cooling techniques showed an increase in power for each PV panel with different increased values. This increase in power showed a remarkable

increase financially when compared to the standard PV. Economic and environmental analyses were conducted on a PV panel with an area of 0.218 m^2.

The governing equations used in the economic study are presented as follows.

$$E = I \times A_{effective} \times \eta_{electrical} \times 30 \quad (9)$$

where

E: The energy produced by the PV panel in kWh.
I: The average solar insolation per day in $\frac{kWh}{m^2 \times day}$.
$A_{effective}$: The effective area of the PV panel in m^2.
$\eta_{electrical}$: The electrical efficiency of the PV panel.

The relative efficiency is the relative difference between the cooled PV efficiency and the standard PV. This relative efficiency is used to calculate the absolute electrical efficiency of each cooled PV case and is quantified as:

$$\eta_{relative} = \frac{\eta_2 - \eta_1}{\eta_1} \times 100 \quad (10)$$

where

η_2: The efficiency of cooled PV (%).
η_1: The efficiency of standard PV (%).
Savings were quantified as:

$$Savings = E \times S \quad (11)$$

where

E: The energy produced by the PV panel in kWh.
S: The price of each kWh.

5.1. Water Cooling

The sun hours vary according to the months of the year. Figure 11 shows the variation in the sun hours with respect to the months in Lebanon. As shown in the following figure, July month reached the maximum of 438.2 h.

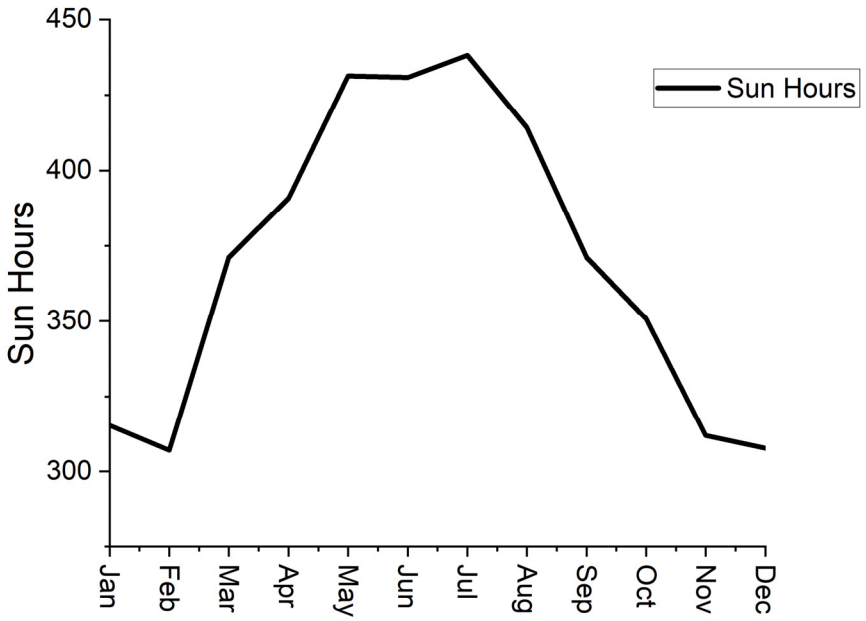


Figure 11. Number of sun hours versus months.

The solar insolation in Beirut, Lebanon is shown in Figure 12 [121].

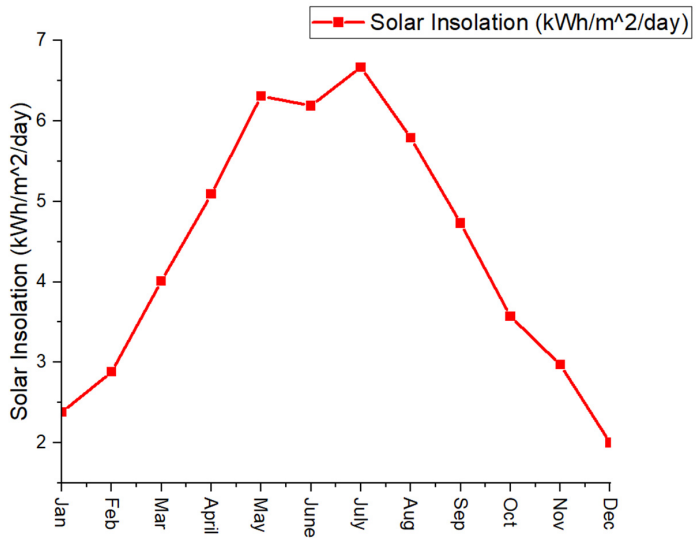


Figure 12. Solar insolation in Beirut, Lebanon [121].

As shown in Figure 12, the minimum solar insolation was recorded in December, with a value of 2 $\frac{kWh}{m^2}$ per day. The highest solar insolation was recorded in July, with a value of 6.67 $\frac{kWh}{m^2}$ per day.

Water cooling methods were found to be effective in cooling the PV panels. As shown in Figure 13, flowing water on the surface of the PV panel was found to produce the maximum energy, with an average of 32.29 kWh compared to the other cooling methods. Following this method, the liquid immersion method, with an average of 32.17 kWh, proved to be the next best. Also, the heat pipe cooling system recorded an average of 31 kWh, while the automotive radiator system recorded the least energy between the cooling methods, with an average of 30.55 kWh. The standard PV panel recorded an average of 29.24 kWh.

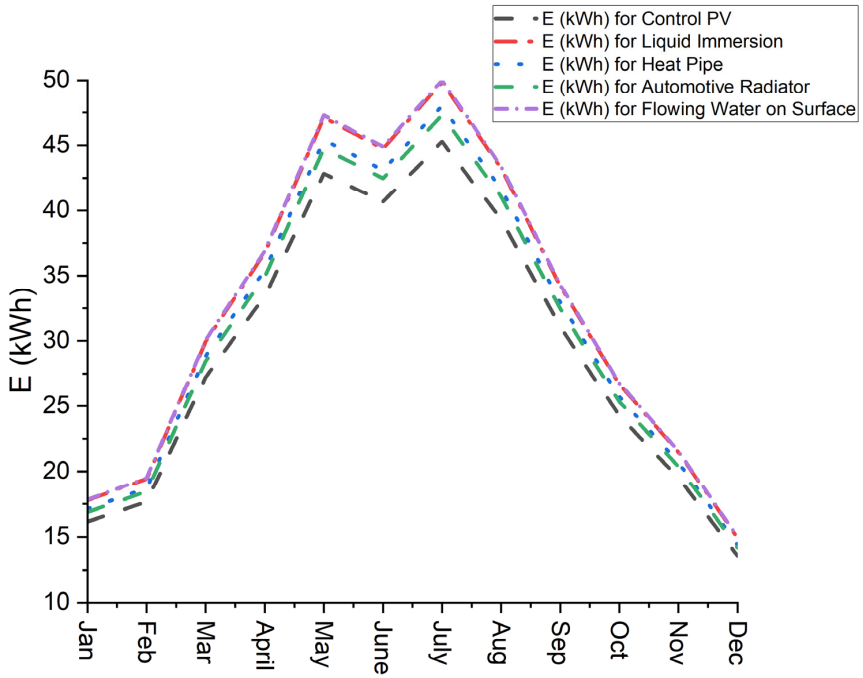


Figure 13. Energy produced versus months for water cooling methods.

Figure 14 shows that the maximum cost saving by the cooling methods was recorded for flowing water on the surface cooling method, with an average cost saving of United

States Dollar (USD) 0.273. The liquid immersion method follows, with an average cost saving of USD 0.263, and the heat pipe cooling method showed an average cost saving of USD 0.157. The automotive radiator cooling method showed the lowest average cost saving, as shown in the following figure with an average compared to a standard PV panel of USD 0.117.

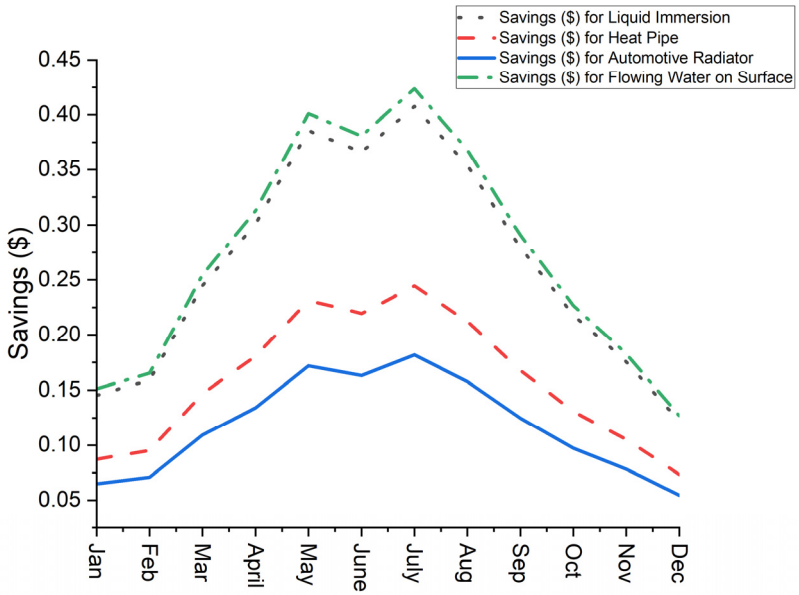


Figure 14. Cost savings versus months for water cooling methods.

5.2. Air Cooling

In air cooling methods, the exhaust air cooling method was found to produce the maximum energy output, with an average of 32.201 kWh. Figure 15 shows the variation in energy produced for air cooling methods with respect to the months of the year.

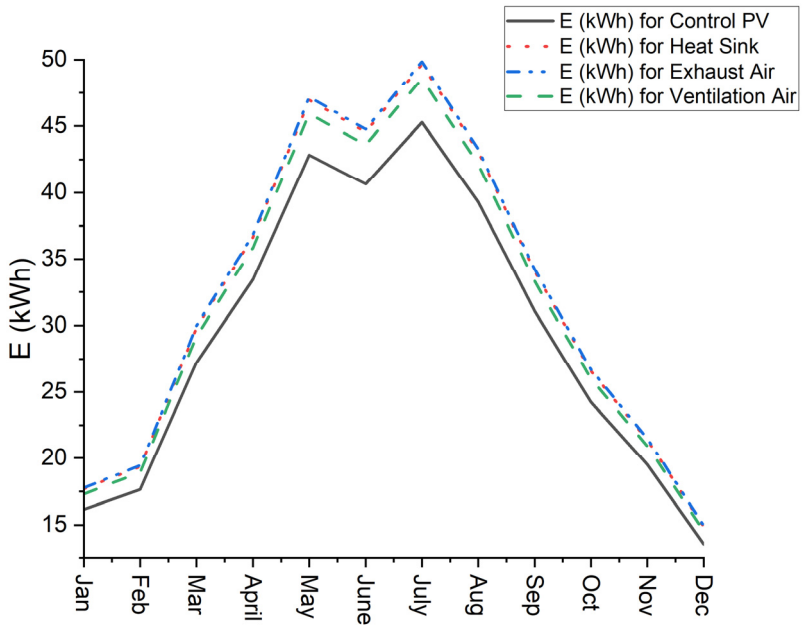


Figure 15. Energy produced versus months for air cooling methods.

As shown in Figure 16, the exhaust air cooling method showed the highest cost savings, with an average of USD 0.265.

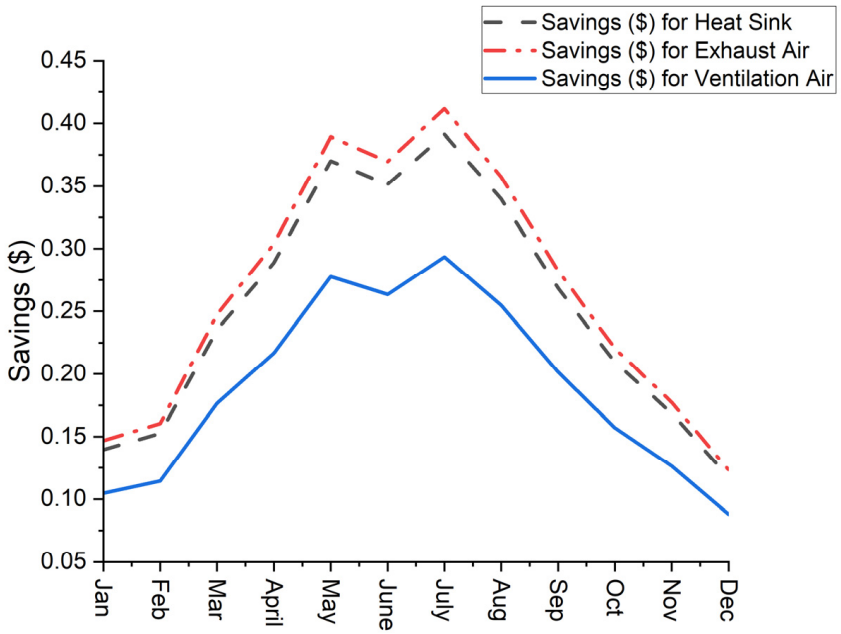


Figure 16. Cost savings versus months for air cooling methods.

5.3. PCM Cooling

Figure 17 shows that cooling by PCM increased the total energy produced compared to the standard PV panel. An average of 31.733 kWh was recorded for cooling by PCM compared to an average of 37 kWh recorded for the standard PV panel.

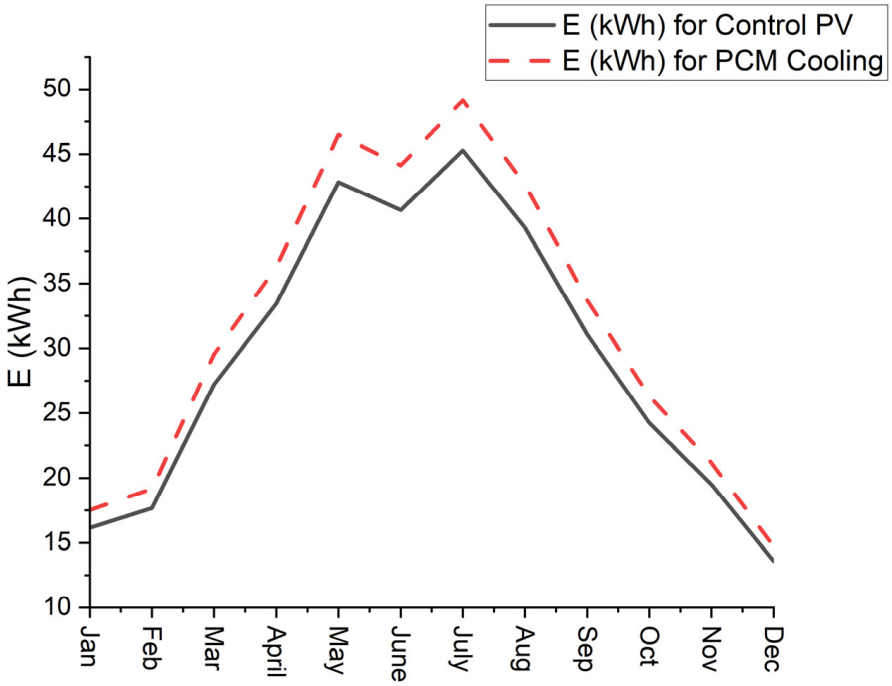


Figure 17. Energy produced versus months for PCM cooling method.

Cooling by PCM increased cost savings compared to the standard PV panel. Figure 18 shows that the maximum amount of money saved by PCM cooling was USD 0.223.

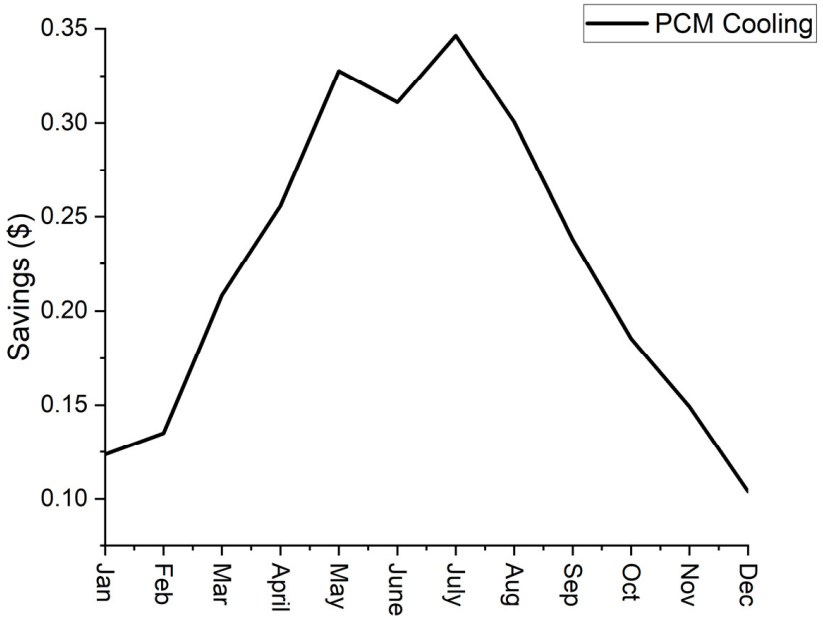


Figure 18. Cost savings versus months for PCM cooling method.

5.4. Other Cooling Methods

As shown in Figure 19, the thermoelectric cooling method was found to produce the maximum energy, with an average of 34.512 kWh.

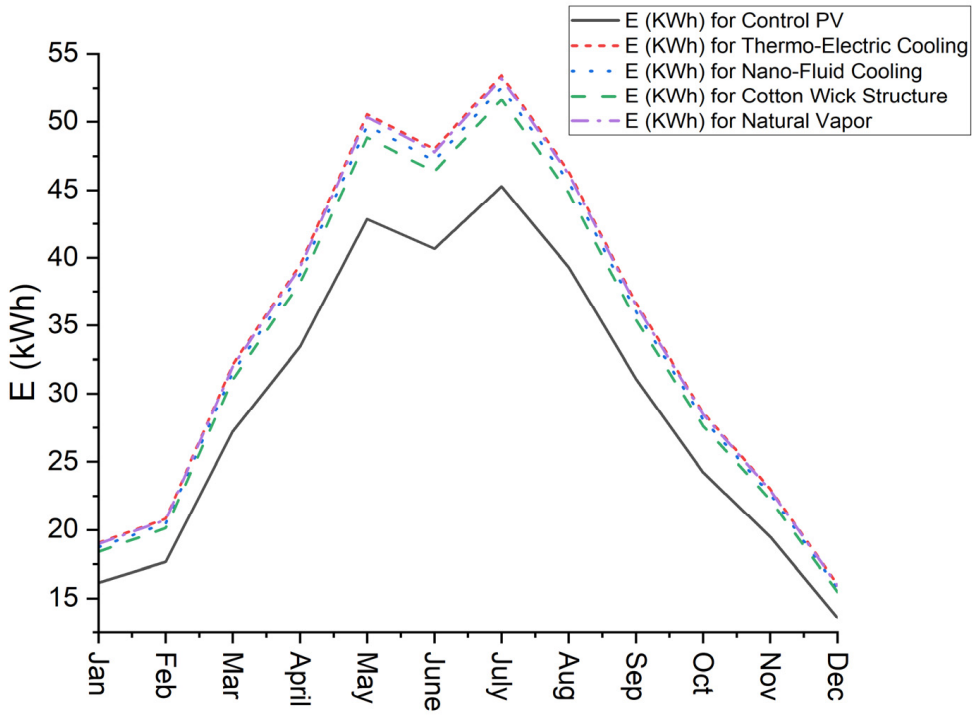


Figure 19. Energy produced versus months for other cooling methods.

The maximum cost saving was recorded by the thermoelectric cooling method, with an average of USD 0.473 recorded in July, as shown in Figure 20.

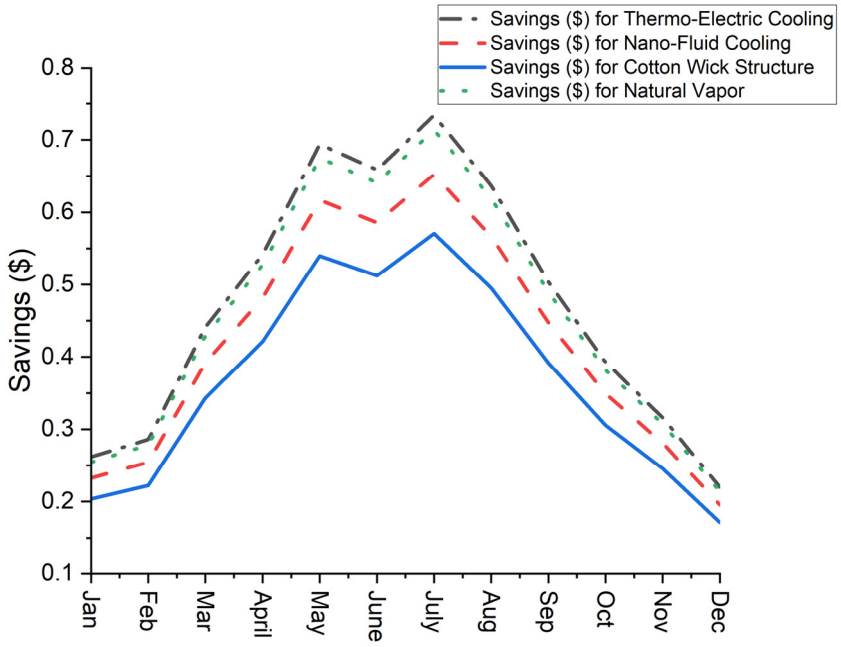


Figure 20. Cost savings versus months for other cooling methods.

In the realm of photovoltaic panel cooling methods, the economic evaluation highlighted the significant benefits of these technologies, both in terms of increased energy production and cost savings compared to standard PV systems. Water-based cooling methods, exemplified by flowing water on the PV panel, have exhibited the highest energy production, yielding an average of 32.29 kWh. This translated into significant financial gains, with cost savings averaging USD 0.273. Liquid immersion and heat pipe cooling systems also demonstrated promising results, while automotive radiator-based cooling methods exhibited slightly lower energy gains and cost savings. In the air cooling category, exhaust air cooling proved to be the most effective, generating an average of 32.201 kWh and yielding the highest cost savings of USD 0.265. Additionally, phase-change material cooling strategies contributed to increased energy production, with an average of 31.73 kWh, resulting in notable cost savings of up to USD 0.223. Among various cooling methods, thermoelectric cooling emerged as the leader in energy production, delivering an average of 34.512 kWh and recording the highest cost savings—particularly in July, with an average of USD 0.473. These results confirm the economic feasibility and financial advantages of applying advanced cooling technologies in PV panel systems, enhancing their ability to drive sustainable and cost-effective energy solutions.

6. Environmental Study

The increased use of fossil fuels has increased CO_2 emissions, which pollutes the air and leads to many serious problems, mainly global warming. Photovoltaic panels were found to reduce CO_2 emissions to the atmosphere as a renewable energy resource.

The governing equations used in the environmental study are as follows.

The CO_2 reduction value is quantified as:

$$CO_{2\ reduced} = E \times P \tag{12}$$

where

$CO_{2\ reduced}$: The amount of CO_2 reduced in kg.

E: Energy produced by a PV panel in kWh.

P: The amount of CO_2 produced per 1 kWh of electricity $\frac{\text{kg}}{\text{kWh}}$.

6.1. Water Cooling

Water cooling methods had a major impact on cooling techniques in reducing CO_2 emissions as a renewable energy resource. A system of nozzles flowing water on the surface of the PV panel was found to result in the maximum CO_2 reduction of 26.509 kg with respect to the other water cooling methods, as shown in Figure 21.

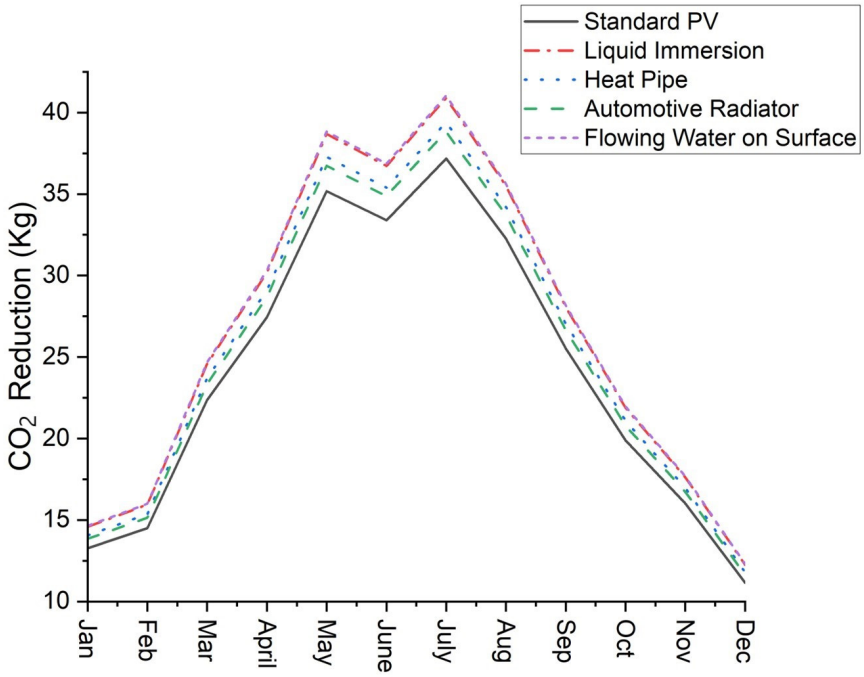


Figure 21. CO_2 emission reduction versus month for water cooling methods.

6.2. Air Cooling

The air cooling technique CO_2 emission reduction varies between the methods. However, as shown in Figure 22, the exhaust air system had the maximum CO_2 emission reduction, with an average of 26.437 kg, compared the other methods.

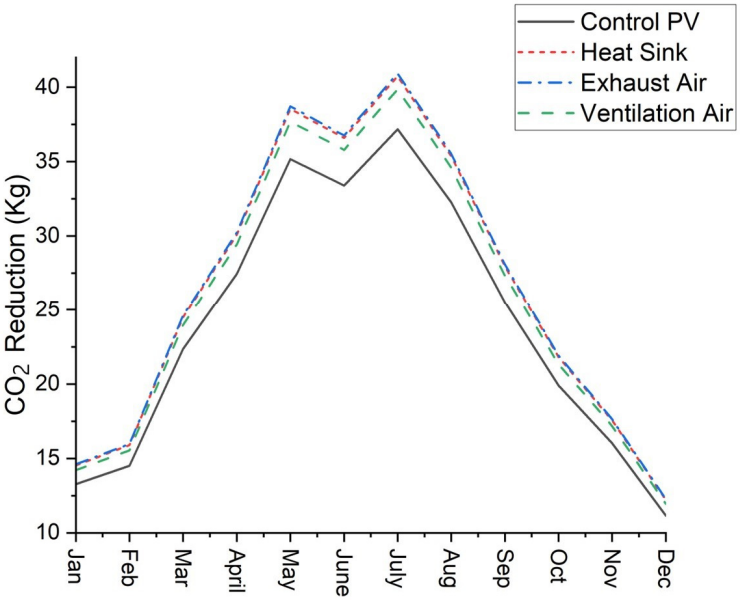


Figure 22. CO_2 emission reduction versus months for air cooling methods.

6.3. PCM Cooling

The PCM cooling method was found to reduce CO_2 emissions by an average of 26.053 kg, as shown in Figure 23.

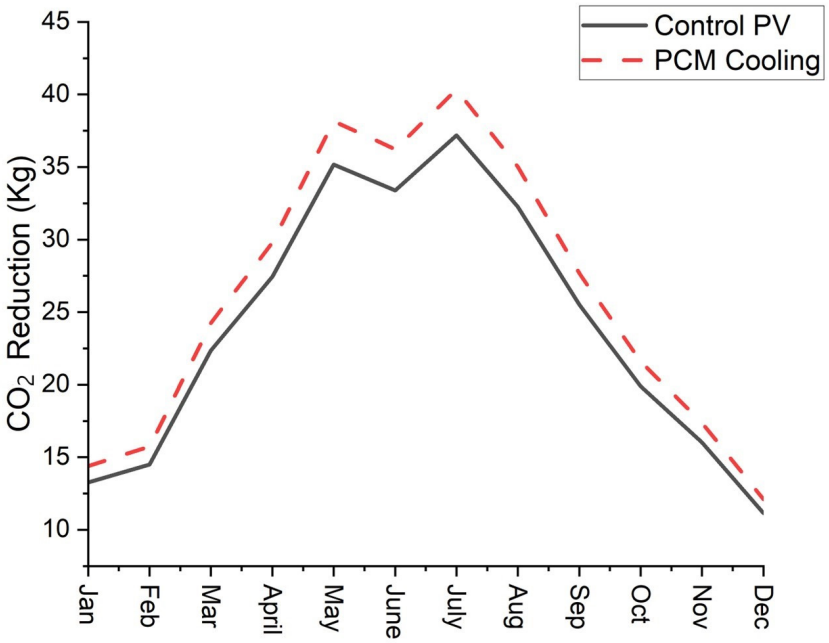


Figure 23. CO_2 emission reduction versus months for PCM cooling method.

6.4. Other Cooling Methods

Other uncategorized cooling techniques had a good impact on the reduction in CO_2 emissions. The thermoelectric cooling system had a maximum reduction in CO_2 emissions, with an average of 28.334 kg, as shown in Figure 24.

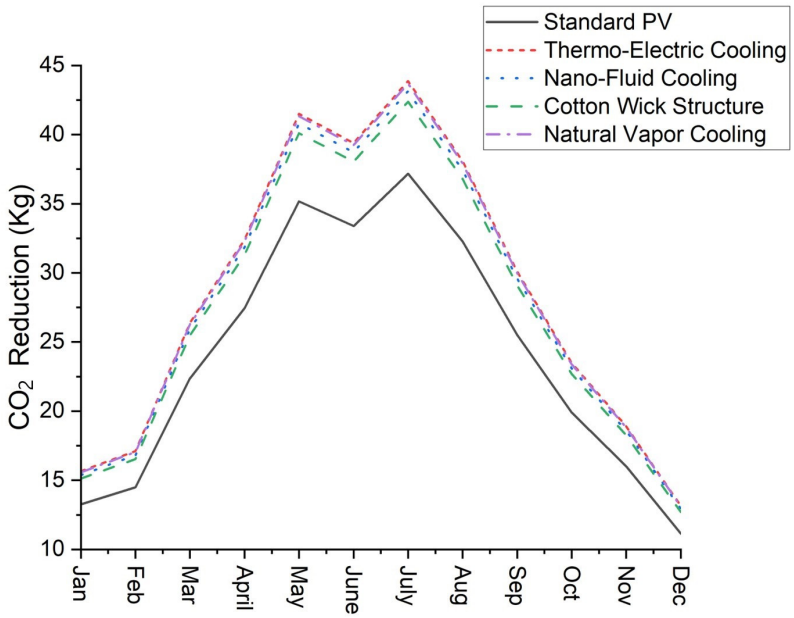


Figure 24. CO_2 emission reduction versus months for other cooling methods.

In short, the escalating use of fossil fuels has led to an alarming rise in carbon dioxide emissions, which has greatly contributed to worsening environmental issues such as global warming. Photovoltaic panels have emerged as a renewable energy resource with the poten-

tial to mitigate these emissions. This study investigated different cooling technologies and their effectiveness in reducing carbon dioxide emissions. Of these, water cooling methods, particularly the nozzle-based system, showed the greatest impact, reducing emissions by 26.509 kg. Air cooling technologies, especially the exhaust air system, also played a decisive role, achieving an average reduction of 26.437 kg. PCM cooling methods contributed to an average weight reduction of 26.053 kg. Even other unclassified cooling technologies, such as the thermoelectric cooling system, succeeded in reducing CO_2 emissions, with an average reduction of 28.334 kg. These results underscore the pivotal role of photovoltaic panels not only in generating renewable energy but also in combating carbon dioxide emissions. As the world grapples with the necessity of tackling climate change, innovative cooling strategies offer promising ways to reduce carbon dioxide emissions, thus promoting a more sustainable and environmentally conscious future.

7. Payback Period

The payback period of each system was studied as investments in order to reveal how much each system approximately costs and how much time it would need to pay the initial investment.

The *payback period* is quantified by the following equation.

$$Payback\ period = \frac{Income}{Cost} \tag{13}$$

where

Income: Profit produced by the system.

Cost: Initial cost of the system.

Figure 25 shows the payback period for the systems consisting of one PV panel each. As shown in the following figure, the automotive radiator system needs approximately 7.576 years in order to pay the initial investment while the standard PV needs approximately 1.9 years.

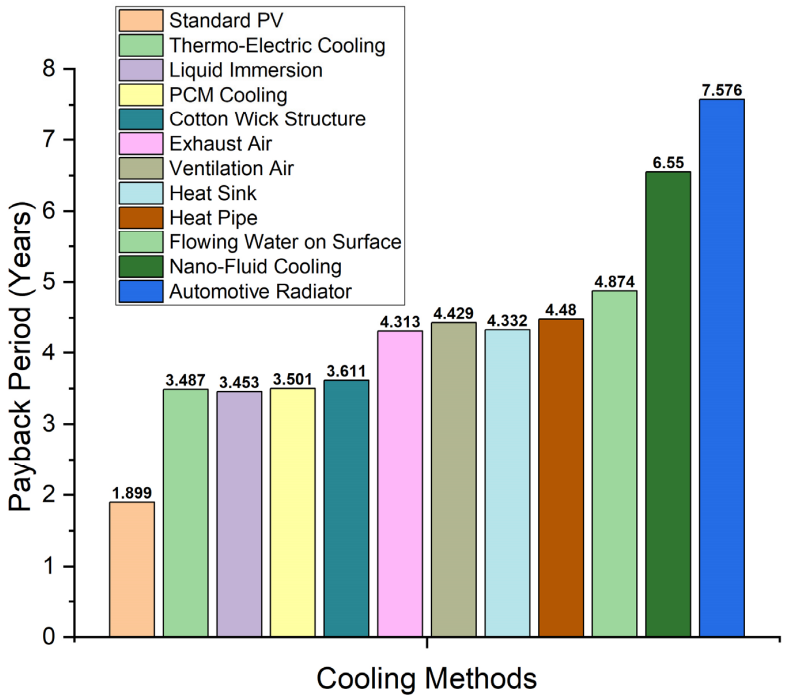


Figure 25. Payback period of each system.

As shown in Figure 25, the return on invest period differs drastically based on the initial investment paid for each system. The standard PV needs approximately 1.9 years to

return the initial investment, while the automotive radiator and nanofluid cooling systems need approximately 7.576 years to return the initial investment paid from their enhanced electric output.

It is true that the payback period has increased when constructing a cooling technique for the PV panel; however, the benefits of the cooling technique on the PV are far more beneficial. The PV panel lifespan increases whenever a cooling system is used because a cooling system decreases its temperature with time. The increase in green energy produced by the PV panel with a cooling system could benefit the environment and be a smart investment on bigger systems, where in the case of cooling, the system needs fewer PV panels to operate and produce higher power outputs, while contributing with a decrease in CO_2 emissions.

8. Conclusions and Recommendations

This review paper addresses the importance of effective cooling strategies to enhance the efficiency of photovoltaic panels. It highlights the negative impact of high temperatures on the performance of photovoltaic panels and emphasizes the necessity of efficient cooling technologies. This review thoroughly explores and discusses a variety of cooling methods, including traditional methods such as water and air cooling, along with innovative solutions such as incorporating phase-change materials, thermoelectric cooling, heat pipes, evaporative cooling, and nanofluids. Furthermore, this review takes into account environmental and economic factors to comprehensively assess the impact of cooling on the performance of photovoltaic panels.

Additionally, the findings of this review emphasize that all evaluated cooling methods have the potential to improve the electrical efficiency of PV panels. However, specific techniques stand out for their superior performance. Notably, among these approaches, the automatic water spraying system, exhaust ventilated air, phase-change materials, and thermoelectric cooling methods exhibited the highest energy production levels. In terms of cost-effectiveness, thermoelectric cooling outperformed evaporative cooling, water-nanofluid cooling, and the automatic spraying system. Furthermore, thermoelectric cooling, evaporative cooling, exhaust-ventilated air, and automatic water spraying demonstrated the greatest reductions in CO_2 emissions.

Moreover, the evaporative cooling technique, along with thermoelectric and PCM cooling methods, showed the shortest payback period. Consequently, evaporative and thermoelectric cooling emerge as particularly promising choices, offering substantial energy improvements, positive environmental effects, and favorable returns on investment. These results emphasize the importance of integrating cooling strategies to improve the efficiency of photovoltaic panels and to maximize the generation of eco-friendly electricity. Given the essential role of renewable energy in addressing climate change and the transition to sustainable energy systems, the integration of efficient cooling technologies can contribute significantly to the advancement of the renewable energy sector.

The development of a highly conductive phase-change material would increase the effect of PCM cooling, enhancing the efficiency and performance of the PV panel. Studies should be targeted on testing different combinations of PCM with other materials, and on the PCM itself to reach a formula where the thermal conductivity is as high as possible and the melting point of the PCM is as close as possible to the standard test conditions of the PV panel. Also, as a future recommendation, the period of analysis at the level of the cooling techniques and methods could be for longer periods, meaning that each cooling method should be studied over different periods of time and for longer hours. Moreover, not many review studies combine all of the cooling methods in one paper. This study mentions nearly all of the cooling methods and a parametric investigation was conducted at the level of environmental and economic analysis, a state-of-the art analysis.

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